

TDT4259 - Applied Data Science

Market Appropriate Pricing for Airbnb Listings in Oslo

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1 Introduction and problem definition

The current report encapsulates the project work completed for the course TDT4259 Applied Data Science at the Norwegian University of Science and Technology. Our project focuses on Airbnb's accommodations in Oslo, Norway, and aims to contribute to Airbnb by developing and evaluating a feature that helps customers determine if listings are priced appropriately. We utilized data science techniques to realize the desired objectives and followed the CRISP-DM framework to ensure business valuable development and data-driven decision-making and recommendations.

1.1 Airbnb

Vacation accommodation has traditionally been offered by businesses that rent out rooms or apartments in their owned building complexes. Therefore, you likely needed to own a building complex, design it for rental, hire staff, and run a profitable operation in order to take part in the short-term rental business. In recent years, however, this norm has been challenged by the likes of Airbnb. The introduction of Airbnb now allows individuals to rent out a part of, or their entire, house or apartment, marking a significant shift in the accommodation market.

Airbnb was founded in the US in 2008 and now operates an online marketplace for short and long term accommodation and experiences in multiple countries worldwide (Wikipedia, 2024). Hosts can publish listings that include information of the property like photos, pricing and amenities. Guests can search through different listings in the area they are visiting and book their desired stay through the platform. Airbnb facilitates the transaction through their platform and takes a commission on every booking. Today, the company has over 5 million hosts and has hosted over 2 billion guest arrivals (Airbnb, 2024).

1.2 Domain and context

A key objective for Airbnb is to make it as easy and enjoyable as possible for both guests and hosts to use their platform. This presents an interesting case where the company not only has to build demand, but in parallel build supply of accommodations that are listed on their platform. Therefore, in order to achieve the desired growth, Airbnb would want to make the experience for hosts and guests as friction-less as possible. After considering both alternatives, we decided to focus on improving the experience for guests, rather than for hosts.

When building guest demand it can be valuable to examine the customer journey, which highlights how guests interact with Airbnb's platform and their experience throughout. The customer journey for Airbnb guests can be broken down into the topics: desire to travel, choice of destination, search for accommodation at destination, choice of accommodation, make the travel to the destination, get access to the accommodation, live in the accommodation, and lastly leave the accommodation. Airbnb have worked with this customer journey and created a well rounded experience for guests through the entire process.

However, through interacting with Airbnb's platform and personally having used the service, we identified a specific pain point in the search for and choice of accommodation part of the customer journey. We struggled with determining if listings were priced at market appropriate rates. Today, guests can see the current price of all listings in the area they are viewing, and after selecting a specific listing they can also see the attributes of the property. Guests are, however, not provided with any information on the pricing history or appropriateness of the price based on the listing's features. To further complicate the situation, guests are often not acquainted with the area they are visiting and do, therefore, not have the local insight to be able to determine if the listings are appropriately priced. Furthermore, accommodation pricing is based on a plethora of features like location, amenities, neighbor pricing, and reviews. This makes it difficult for individual guests to analyze both the price and features for all the listings they are interested in. Hence, finding out if an accommodation is priced appropriately in relation to the market becomes a near impossible task.

Seeing that this issue could affect the customer experience, there is a business incentive to examine the problem further. It would be beneficial to determine if something could be done to increase the customers ability to evaluate listing prices.

1.3 Problem definition

Guests can, as mentioned, have trouble determining if accommodations that they are interested in are appropriately priced. To alleviate this issue and improve Airbnb's customer experience we believe that a data science approach, utilizing public Airbnb listing data, can be effective. Based on this, we formulate the following problem definition:

Problem definition: Can data science be utilized to sufficiently suggest if an Airbnb listing is appropriately priced based on features like location, amenities, reviews and neighboring pricing? Here sufficiently will be determined by model evaluation metrics that will be described in Section 3.3.4.

1.3.1 In-scope

We choose to focus its effort on a single geographical region, due to the belief that regional patterns could be prominent and generalization between different cities could, thus, be challenging. Still, we believe that the methods used can be replicated successfully for other cities and the results shown can be used to justify further investigation into the problem. After considering multiple cities to focus on, we chose to examine Oslo, Norway, due to its relation to the study country of the project group members.

1.3.2 Out-of-scope

The current project will not examine other locations than Oslo.

1.4 Team roles and responsibilities

The project team consists of seven members, all university students with different engineering and national backgrounds. No distinct roles, like project manager or developer, were determined. All project members took ownership of the project progress and meetings were managed by different members on initiative. Furthermore, all members expressed a desire to work on the development of the technical aspects of the project. However, the main responsibility for the technical development was led by members with background in data science, machine learning and computer science. Report writing was mainly led by members with non-coding backgrounds, but all members contributed to the writing of the report. The members and their respective backgrounds and responsibilities in the project can be seen in Table 1.

Table 1: Team members' background and project responsibilities.

Name	Background	Responsibilities
Alexander Sundquist	Industrial Engineering and Management	Report writing
Berk Sen	Information Systems	Data analysis, data cleaning, and modeling
Cas Weijers	Operations Management and Logistics	Feature engineering, and modeling
Nicolas Fröhlich	Social and Economic Data Science	Data analysis
Niklas Olesen	Computer Science	Data analysis, feature understanding, and modeling.
Lukas af Geijerstam	Biotechnology and Bioinformatics	Report writing
Timon Selnes	Software Engineering	Implementation

2 Background

The following sections describe the objective of the project and how it relates to Airbnb's business goals. Additionally, the data science project management and data strategy used in the project will be presented.

2.1 Project objectives

After gaining an understanding of Airbnb's business and formulating the problem definition in Section 1.3 the objectives of the current project were developed. The project objectives and how they relate to the problem definition and Airbnb's business objectives will be described below.

2.1.1 Objective 1: Determine how data science can be used to suggest if listing prices are appropriate

First, to tackle the problem definition, a natural first step of investigation is to determine how data science can be used to suggest if listing prices are appropriate. Answering this would provide the foundation for developing a functional solution that could be utilized by Airbnb. Therefore, it is both crucial for the project and for realizing Airbnb's business objective.

2.1.2 Objective 2: Develop a functional first version of the proposed data science solution

Second, to evaluate if data science could be used to suggest if listings are appropriately priced, the proposed solution from Objective 1 should be developed. A mere theoretical approach would be insufficient for the project objectives and to provide true business value for Airbnb a minimum viable product would be necessary.

2.1.3 Objective 3: Determine if the proposed solution performs sufficiently in suggesting appropriate listing prices and fulfills Airbnb's business objectives

Lastly, to answer the part of the problem definition that specifies if data science can *sufficiently* suggest appropriate prices, the performance of the solution must be included as an objective. Simply building out a functional iteration of the proposed solution would provide an idea of if the solution could satisfy the business need. But the solution must be able to perform to the level required in shipped features. The project must therefore show the required performance in the prototype developed in Objective 2 or be able to suggest that the performance could be improved to meet the requirements. Failing in doing so, the business value for Airbnb would not be realized.

2.2 Data science project management and data strategy

2.2.1 CRISP-DM

The method used in the project to answer the problem statement and achieve the project objectives is based on the industry best practice framework CRISP-DM. CRISP-DM stands for CRoss Industry Standard Process for Data Mining, and it is a widely used workflow for machine learning and data mining projects. It has a general structure that allows using it for different types of projects across different industries, with the contents of each step of course adapted for the problem at hand.

The structure of the CRISP-DM workflow is shown in Figure 1, and the steps that will be taken for this project are presented below. The description as well as the aforementioned figure is largely based on the writings of Balo et al. (2018).

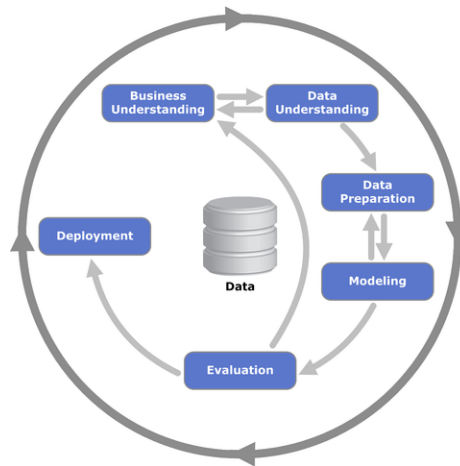


Figure 1: General structure of the CRISP-DM workflow.

Business understanding

The first step of the CRISP-DM workflow is to gain an understanding of the business the project is designed for. This includes both a general view of the business and its relevant business field, and a more detailed understanding of the problem that is to be solved. Without this type of background knowledge, developing a model becomes both difficult and unproductive.

For this project, the business understanding will mainly come from examining official Airbnb documentation and prior knowledge about the company and its business model. This is considered sufficient for this project, but if the product was to be implemented by Airbnb, additional information about the company and additional specification of the project would be needed.

A crucial first step in business understanding is defining the business problem. This should be done in contact with relevant employees at Airbnb, such as subject matter experts (SMEs), data analysts and people working on the business side. For the case of this report, the business problem is formulated as the problem definition in Section 1.3.

When this is done, a common next step is to analyze the current state in terms of available resources, personnel and data. For our project, we have no actual connection to Airbnb and will thus only be able to do this regarding the dataset we will be using. Another part of this step is to discuss success criteria and business objectives with relevant people - no such discussions will be made in this instance either. It is however worth noting that this would need to be carried out prior to performing the later parts of the project, were this to be implemented.

Balo et al. (2018) describes "defining data mining problem" as the next step of business understanding. This is seen as a pre-analytical step of understanding the data, to enable more proper data understanding later. In our case, this simply comes down to choosing and having a quick look at the dataset, and starting to think about what types of methods

could be used to solve the problem. In our case, this simply comes down to choosing and briefly examining a dataset. It is heavily related to objective 1 stated in Section 2.1, which is to determine if data science can be used to tackle the problem at hand. In a real business scenario, this step is carried out in contact with both data analysts/scientist and business stakeholders, to make sure that the problem has a viable solution.

When all of these steps are finished, a natural procedure is to write a project plan. This is a detailed document covering business objectives, success criteria, data analysis and machine learning workflows to be followed, as well as a plan of all the steps in CRISP-DM with estimated timelines and risks. This is an important document in most projects, since it provides a clear view of the tasks at hand, both for other relevant people to give their opinions and to be referred to as the project progresses. No such extensive document is written for this project. Instead, the relevant information is presented under the current section.

Data understanding

After gaining an understanding of the business side of the problem, it is crucial to examine the data the project will be using. This includes looking at the types of features - are they categorical or numerical? How many features are there in total, and are there any that could be disregarded before model development? Are there any missing values? Questions like these need to be assessed and answered properly before any actual model development can take place. Otherwise the quality of the model will be severely lowered, and one would also not understand the predictions that are produced.

Another important part of data understanding is to look at the relationships between features, for example by creating a matrix showing all correlation coefficients. This step can also include plotting variables against each other - either those with strong correlation, or variables that are hypothesized to impact the feature to predict. In the case of predicting the price of Airbnb listings, one such feature could be the number of bedrooms in the accommodation.

Balo et al. (2018) describe three distinguished steps of this phase - data collection, data description and exploratory data analysis. The data collection phase includes choosing the appropriate data from what is available. For a customer with a large data bank, this step is a crucial first step in narrowing the scope of the project, since the model will be based on the used dataset(s). For this project, an openly available dataset from Airbnb listings in Oslo for the period 29/3 - 29/6 2024 will be used, downloaded as one large csv file.

Data description is a necessary following step, in which basic information about the dataset is produced and analyzed. This means finding out the volume, number of features, relevant units, what features might be important for the project etc. In addition, some descriptive statistics such as mean, median and variance of features might be calculated.

To gain a detailed understanding of the data, a thorough exploratory data analysis (EDA) is carried out. This involves visualizations, finding possible correlations, and identifying outliers and missing values. Data description can be seen as a starting point

of the EDA, and both of these are presented in section 3.1.1. Balo et al. further define data quality analysis as separate from the EDA, and in this step missing values, wrong metadata etc. are noticed and analyzed. This is also part of the EDA section in this report.

Data preparation

In this next phase of the CRISP-DM cycle, findings from the EDA are used to modify the dataset to facilitate model development. This includes removing or imputing missing values, formatting values concisely, and perhaps removing outliers. Feature selection is also a part of this phase, and it is separated from the former, which Balo et al. call "data wrangling". Feature selection is part of feature engineering, and the broader category not only involves selecting appropriate features based on hypotheses and/or importance analyses, but also normalization, one-hot encoding of categorical variables etc.

In this project, feature selection is done both based on what features are likely to affect the price of a listing, as well as an analysis of their importance, further detailed in 3.2.2. Other data preparation steps are described in 3.2.1.

It is important to note that data preparation is a time and resource heavy step of any machine learning project, and this project is no exception. To ensure optimal model performance, high quality data is necessary. In order to understand how different data modifications affect the developed models, this entire phase will be carried out partially in parallel with the next phase - modeling.

Modeling

Modeling can be seen as the main phase in a machine learning project, since this is where the actual product is built. However, it relies on proper understanding and data preparation in the former step to be effective. The first step in modeling is to select modeling techniques. Different techniques are normally tried, since all techniques perform differently on different datasets. It is often not clear which model will have the best performance for a certain project, although hypotheses about this can be made. These are based on things such as data quality, performance requirements and business goals - all things which should be defined before modeling takes place.

When the models have been selected, they are trained on the processed dataset. Training involves splitting the dataset between training and testing data, where the latter is unseen for the entire training process. It is important to note down information about what models are used with which parameters, and what results they produce, for later evaluation steps.

All models have different parameters that affect their performance. In order to choose these appropriately, they are tuned during the modeling phase. Tuning is normally done using some form of cross-validation, where the training set is split into several parts where a few are used for training and a few for validation. This is repeated for different subsets to avoid overfitting. This approach will be taken for training all models in this project, and is further detailed in

It is important to choose reasonable metrics in line with the aim of the project to assess the quality of different hyperparameter choices. These can be related to things such as accuracy, robustness or speed. When these have been taken into account and the models have been developed, the models are compared and the best performing model is chosen. Performance should be based not only on the accuracy of the models, but on all aspects related to the overall quality of the final product.

Evaluation

During evaluation the main objective is to assess the degree to which the chosen model meets the business objectives. If time and budget allows, testing the model on test cases can also be done here. It is also recommended that a thorough review process is conducted to identify if any important task or factor has been overlooked by during the project. Lastly, a determination of next steps should be done where reasons for future actions of continuing to deployment, iterate further, or initiate a new project. (Dalal, 2024)

Deployment

When a final model has been evaluated and deemed as satisfactory, it has to be implemented in an effective manner. This is referred to as Deployment, which is the last step in the CRISP-DM workflow. Considerations during this phase include potential implementation issues based on available software, hardware, resources etc. Recommendations for monitoring and maintaining the system are also important, as well as thinking about potential future risks. These points are discussed in Sections 5 and 6.

2.2.2 Design thinking

To complement the CRISP-DM framework we used core design thinking principles throughout the project. Norman (2013), describes design thinking with the following words:

”Designers resist the temptation to jump immediately to a solution for the stated problem. Instead, they first spend time determining what basic, fundamental (root) issue needs to be addressed. They don’t try to search for a solution until they have determined the real problem, and even then, instead of solving that problem, they stop to consider a wide range of potential solutions. Only then will they finally converge upon their proposal. This process is called design thinking.” (Norman, 2013. p. 219).

The process of design thinking is at its core about solving the right problem, not solving the wrong problem right. We utilized this idea when researching the project. After we chose to focus on Airbnb, we realized that we had all struggled with determining if prices of listings were appropriate. This indicated that this was indeed a real problem that could be solved. Only then did we start exploring solutions. To ensure that we exhaustively considered a range of potential solutions, defining how the problem could be solved was even included as a separate objective for the project, see Section 2.1.1 Objective 1. In using the core principles of design thinking, the likelihood of the project results having a real world impact were greatly improved.

2.3 Relevant examples

For structuring the report and for the general workflow of the project, we took inspiration from the project report "Examining Regulatory Impact on Hosting Practices in New York - Airbnb" from an earlier iteration of this course (Døskelad et al., 2023). This project also used open datasets from Airbnb, but with the goal of understanding how regulations and other market events had effected the Airbnb listings in New York. Although their project had a different type of goal, it provided valuable insights and inspiration for handling the dataset and structuring our work.

Other than that, most of the ideas for the project came from our earlier experiences, as well as from the background regarding data science projects explained above.

3 Method and analysis

To achieve the project objectives described in Section 2.1 the systematic methodology described in Section 2.2.1 was used.

3.1 Data understanding

The data set in use comes from the website *Inside Airbnb*, which is a 'mission driven project that provides data and advocacy about Airbnb's impact on residential communities' (Cox et al., 2024). It sources its data from publicly available information from the Airbnb site and provides 'verified, cleansed, analyzed and aggregated' data sets free of charge available for download on their website. We chose the most current and most extensive data set ('Detailed Listings data') for Oslo from June 29th, 2024.

3.1.1 Exploratory data analysis

The original data set has a total number of 10,099 observations and 76 features. To get a better understanding of the data, we first went through the detailed text description of the features which is provided by *Inside Airbnb*. This allowed us to apply our domain knowledge to feature selection, which is described in detail in section 3.2.2.

The following paragraphs and figures provide insights into the descriptive statistics of the chosen features, see 3.2.2. First, we examined the percentage of missing values in our relevant features as depicted in Figure 2. Five features, especially `price`, `reviews_per_month`, and `bathrooms`, had some missing value. This was handled with imputation as described in section 3.2.1.

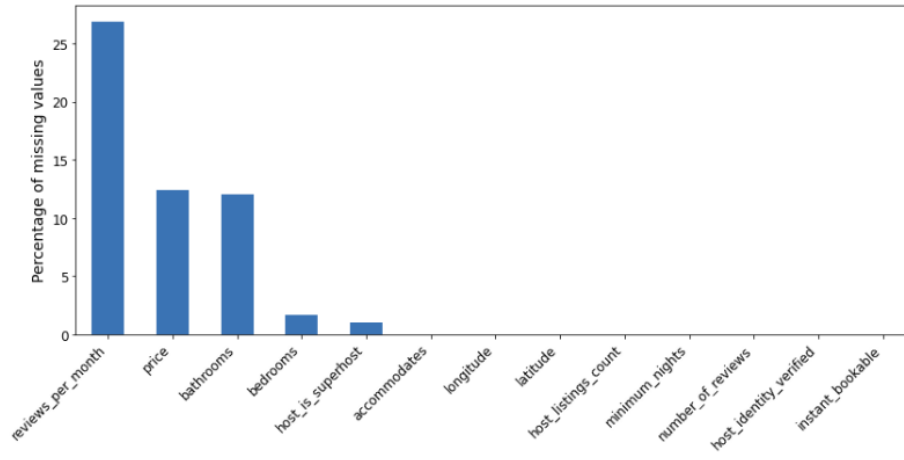


Figure 2: Percentage of missing values for relevant features.

Next, a closer look at the descriptive statistics as shown in Table 2 for each feature of interest pointed us to the skewness of features (if mean and median differ strongly) and to potential outliers (if min/max values lie far from the expected range).

Table 2: Descriptive statistics of listings

	min	mean	median	std	max
reviews_per_month	0.01	1.22	0.66	1.57	13.21
price	160.00	1691.40	1304.50	2297.92	110000.00
bathrooms	0.00	1.14	1.00	0.41	5.00
bedrooms	0.00	1.60	1.00	0.94	10.00
host_is_superhost	0.00	0.17	0.00	0.38	1.00
accommodates	1.00	3.44	3.00	1.81	16.00
longitude	10.59	10.76	10.76	0.04	10.94
latitude	59.82	59.92	59.92	0.02	60.04
host_listings_count	1.00	2.86	1.00	10.05	131.00
minimum_nights	1.00	3.97	2.00	12.25	500.00
number_of_reviews	0.00	16.60	4.00	43.62	1161.00
host_identity_verified	0.00	0.80	1.00	0.40	1.00
instant_bookable	0.00	0.09	0.00	0.29	1.00

The map in Figure 3 shows the spatial distribution of Airbnb prices for one night. To account for the strong right-skewness, we used logarithmic scaling for mapping the values to colors. A large variation in prices over space is apparent from the map, which is a sign for the importance of location as a model feature.

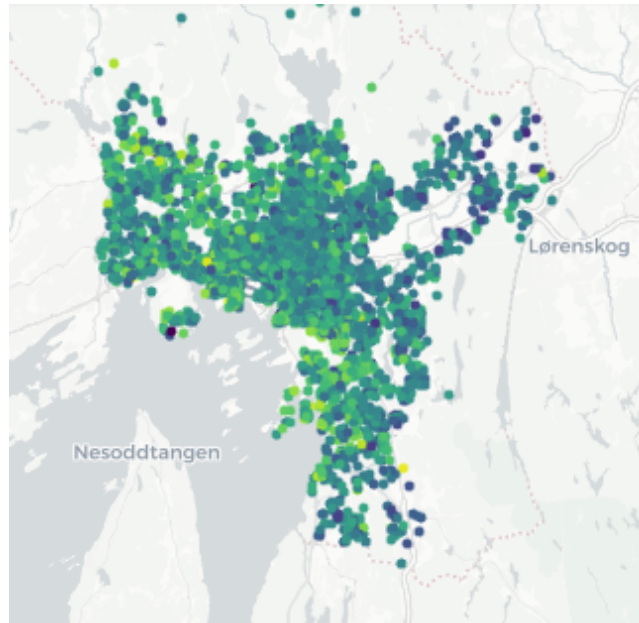


Figure 3: Airbnb price distribution in Oslo.

Note: Brighter color indicates higher price, using logarithmic scaling.

Of the over 10,000 original observations, only 8,850 do have a value for price. The mean price in the full data set is 1,639 NOK, while the median price is 1,304 NOK. This is the result of the strongly left-skewed distribution of the price feature, as is common for money-related data.

Note that the plots presented below are based on data that has already undergone pre-processing as described in 3.2.1.

The scatter plot matrix in Figure 4 exemplarily plots the variables `price`, `bathroom`, and `bedroom` against each other. The plots in the downward diagonal show the distribution of each of the three variables. The remaining six plots (in the bottom left and upper right) show two of the variables plotted against each other. The expected weak trend can be seen for example in the bottom left plot: with a higher number of rooms comes a higher price.

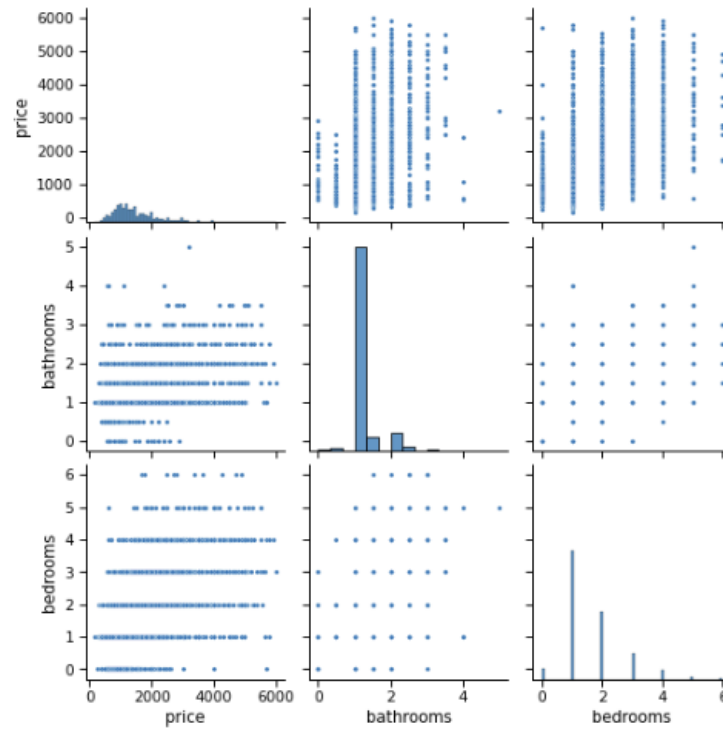


Figure 4: Scatterplot matrix of price, bathroom, and bedroom.

3.2 Data preparation

3.2.1 Preprocessing

In preparation for modeling on Airbnb listings data, a series of preprocessing steps were applied to ensure data consistency, handle missing values, and remove outliers. The following steps were performed:

1. **Price cleaning:** The `price` column was cleaned by removing commas and dollar signs, then converting it to a numeric format to be ready for training.
2. **Boolean conversion:** Certain categorical features were binary (e.g., `host_identity_verified`, `instant_bookable`). These were standardized by converting true/false values to 1 and 0, respectively. Additional checks handled inconsistencies where boolean columns were represented as strings ('t'/'f') instead of booleans.
3. **Missing value imputation:** To handle missing values, numerical columns were imputed with their respective means. Non-numeric columns with missing values were dropped to maintain data quality without distorting statistical features. This approach ensured that the remaining data points in numeric fields were representative of the column's distribution.
4. **Outlier removal:** For the final model training, we only removed outliers from the price feature, since removing other outliers did not improve the performance of the models. To determine which outliers needed to be removed, a distribution was fitted to the uncleaned price data. As can be seen in Figure 5, the data is distributed lognormal, based on the lowest sum of square errors. To remove outliers, the data was first log-transformed to make it normally distributed. After that, values of 3 standard deviations above or below the mean were dropped. The last step was to transform the data back to its original values.
5. **Scaling of independent variables:** To normalise the range of the independent variables, a min-max scaling method is used. This preserves the original distribution of the data and makes it easier for the models to handle.
6. **Scaling of the target variable:** As it was found that the distribution of the target variable price is skewed, it was later decided also to transform this variable using log-transformation. This was done to stabilise the variance of the price values, which enhances the models' ability to capture the underlying patterns in the data.

This last step was crucial for improving model stability and predictive accuracy by reducing the influence of atypical data points. The distribution for each of the features after preprocessing can be seen in Figure 6. From Figure 6, we also see that the distribution for features such as `reviews_per_month`, `minimum_nights` and `number_of_reviews` are very unstable. As part of our experimentation, we also tried removing outliers of different thresholds for these features, but every try worsened our models predictions. Therefore, we only removed outliers for the target feature. Through these preprocessing steps, the data was transformed into a clean and consistent state, ready to be trained upon.

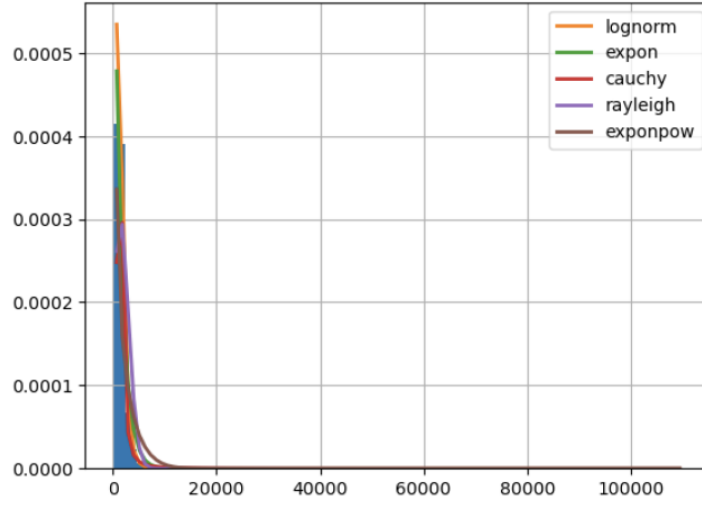


Figure 5: Probability distribution of the price variable.

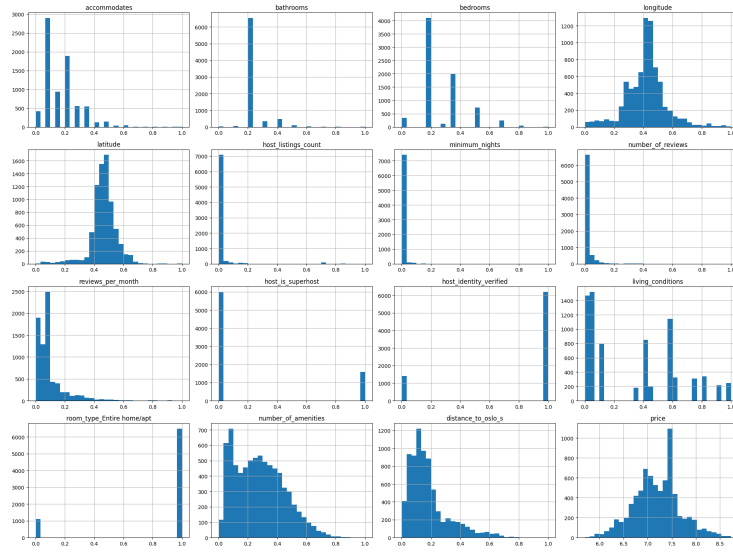


Figure 6: Feature Distribution

3.2.2 Feature engineering

The provided dataset is rather large and not all variables will have an impact on the price prediction. Therefore several tests were done to determine which variables to keep. The first was to use all the variables in an OLS regression and see which would have a significant p-value (≤ 0.05), these were noted down. The second test was to use the permutation importance test for the Random Forest (RF) regressor. This test measures the impact (error) of increasing a variable's value by 1 on the predictions made by the RF. The importance of this variable is then the size of the error made by the model. The higher the absolute error the more predicting power the value has. Last, domain knowledge was used to determine which variables to include in the final model. The selected features and their descriptions can be seen in Table 3.

Table 3: Selected features and their descriptions.

Feature	Description
Accommodates	The number of people that can stay in the Airbnb.
Bathrooms	The number of bathrooms the property has.
Bedrooms	The total amount of bedrooms.
Longitude	The longitude coordinate of the Airbnb location.
Latitude	The latitude coordinate of the Airbnb location.
Host listings count	The number of listings the host has.
Minimum nights	The minimum number of nights a tenant must stay.
Number of reviews	The total number of reviews on the property.
Reviews per month	The mean number of reviews per month a listing receives.
Host is superhost	Describes whether the host is a 'superhost' or not.
Host Identity verified	If the identity of the host is verified by Airbnb.
Room type Entire home/apt	Indicating if the listing is an entire home or apartment.

To get more information out of the dataset and selected variables, feature engineering/extraction was applied. The following new features were created:

- **Number of amenities:** The original data set included a list, mentioning all the amenities included in the Airbnb listing. This was too much and too messy to one-hot encode. To still include it in the analysis, as it likely describes the luxury of the listing to some degree, the number of amenities was calculated. This was done by counting the individual amenities mentioned.
- **The living conditions per neighbourhood:** A study by Akopian et al., 2015 gave the different neighbourhoods of Oslo a score for their living conditions. The study was based on use of narcotic substances, which occurred more towards the city centre than in the outskirts.
- **Distance to Oslo central station:** Most of the users of Airbnbs will be from outside of Oslo, therefore, a good measure of centrality and travel distance after arriving in Oslo, will be the distance from the central train station. This distance is calculated using the geodesic distance.

3.3 Modeling

After the data understanding and processing it was quite clear that a prediction model was best suited to suggest if the price of an Airbnb listing was appropriate. Given the data that is gathered, three different modeling approaches were taken:

- Linear regression
- Random Forest Model
- Multilayer Perceptron Model

The choice for these three models is based on their degree of interpretability, with linear regression being the easiest to understand and the MLP being hard to understand as it is a black box model. These models are explained in the following sections.

3.3.1 Linear regression

For regression, an Ordinary Least Squares (OLS) model is used. This method tries to find the best coefficients for each variable such that the sum of the squared errors, between the fitted line and the actual observation, is minimised. If there is a strictly linear relationship in the data, this model will likely perform well, however it is unable to detect any other patterns.

This method is the easiest to explain of the three proposed and offers good help in deciding whether to include features or not. Besides this, it is also useful to check if the data cleaning methods are not too strict or lenient.

3.3.2 Random Forest Regressor

A Random Forest Regressor is an ensemble learning method that constructs multiple decision trees during training and outputs the mean prediction of the individual trees. This method is particularly effective for regression tasks due to its ability to handle non-linear relationships and interactions between features which makes it more versatile compared to linear models like OLS. The method is also robust, as it is less prone to overfitting compared to individual decision trees, thanks to the averaging of multiple trees.

However, the complexity of Random Forest makes it harder to interpret compared to simpler models like OLS. It could also be computationally intensive, requiring more time and resources for training, especially with large datasets.

3.3.3 Multilayer Perceptron Model

A Multilayer Perceptron (MLP) is a type of feedforward artificial neural network that consists of multiple layers of nodes (neurons) connected in a directed graph. Each node represents a perceptron, which performs computations and transfers information between layers. We selected this model as our more complex option to compare its performance with that of simpler models and observe how it captures relationships in the data.

The MLP was constructed with an input layer, one or more hidden layers, and an output layer. The input layer corresponded to the selected features from the Airbnb listings, including accommodations, amenities, and host characteristics. Hidden layers were added to enable the model to learn complex patterns in the data. The output layer consisted of a single neuron, representing the predicted price.

The model was trained using backpropagation with Mean Squared Error (MSE) as the loss function. During training, weights were adjusted through gradient descent to minimize the difference between predicted and actual prices. To optimize model performance, we performed hyperparameter tuning across multiple parameters, such as hidden layer sizes & depth, learning rate, and regularization strength, to identify the best fit.

3.3.4 Model evaluation

To assess and compare the performance of the models, we focused on two primary metrics: R^2 and root mean square error (RMSE). Additionally, we examined residual plots to visually inspect the quality of the model fits and identify any potential patterns or issues with residuals.

- **R^2 :** This metric provides insight into the proportion of variance in the dependent variable that is explained by the independent variables in the model. Higher R^2 values indicate that a model is better at explaining variability in the data, though it is not solely indicative of predictive accuracy.
- **RMSE:** RMSE measures the average magnitude of the prediction errors. Lower RMSE values indicate better model performance, as they represent smaller differences between observed and predicted values.
- **Residual Plots:** Residual plots were used as a diagnostic tool to visually assess the performance of the models.

All models we tested were quick to train, so computational speed did not influence our choice of the best model.

The choice of the best model was based on maximizing R^2 , minimizing RMSE, and ensuring residuals showed a random distribution. This allowed us to select a model that both explained the data well and provided reliable predictions without obvious patterns in residual errors.

4 Evaluation and interpretation

In the following section the conducted data analysis and the results will be presented. The reliability of the pipeline and the shortcomings of the used method will also be discussed.

4.1 Results

The results that were generated from the method and analysis described in Section 3 will be presented here in relation to the three project objectives.

4.1.1 Objective 1: Determine how data science can be used to suggest if listing prices are appropriate

In line with the design thinking methodology described in Section 2.2.2, after trying to find a real problem to solve we attempted to explore possible solution alternatives before deciding on a solution.

In the beginning, when gaining a data understanding ideas of which data science solutions could solve the issue were continuously thought of. Throughout the process of data understanding and later data processing the solution alternatives crystallized even more. The general consensus within the group was that a machine learning approach made most sense. The problem of determining if listing prices are appropriate could be defined as a prediction task, where a listing's appropriate price could be predicted based on similar listings' pricing. The problem could, thus, be framed as a supervised machine learning regression task.

Throughout the data understanding and data processing phases it became clear that pure data analysis, like visualization, could not sufficiently suggest if prices were appropriate. Visualizations could provide better insight into for example which areas were more expensive, like Figure 3. This could be used by users to determine if a specific listing's price fit into the general trend of the figure. However, as can be seen in Figure 3, pricing trends are ambiguous and often too complex to infer without other assistance. These 'hidden patterns' and the large data volumes make it difficult, and almost impossible, to do the predictions "by hand". Therefore, machine learning, which is adept at finding hidden patterns between features, seemed like a natural way to go for this specific data science problem.

4.1.2 Objective 2: Develop a functional first version of the proposed data science solution

After deciding that machine learning could be used to suggest if listings are appropriately priced, as part of objective 1, a functional machine learning model was developed. Three models, OLS, Random Forest, and Multilayer Perceptron Model, were evaluated and the best performer was chosen as the final model.

OLS: After fitting the model, the test set returned an R^2 of 0.56, indicating an average explanation of the variance in the model, and gave a **RMSE** of 528.3. The residuals can be found in Figure 7. They look to be centred around 0 and randomly distributed. However, because the data is somewhat skewed to the lower prices, most points are located on the left side of the graph.

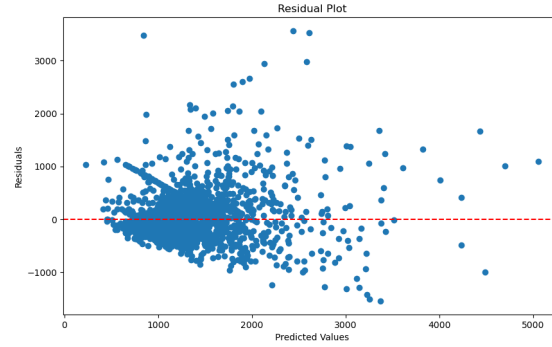


Figure 7: OLS residual plot

Random Forest Regressor: The Random Forest regressor found the best tree within 1000 iterations. Increasing this number to 10,000 or 100,000 did not yield any better-fitting trees. The final value of the R^2 is 0.65 and the test set returned a **RMSE** of 497.9. These results indicate the RF model has a moderately strong fit. The residuals can be seen in Figure 8, where the same observations as for the OLS model can be made.

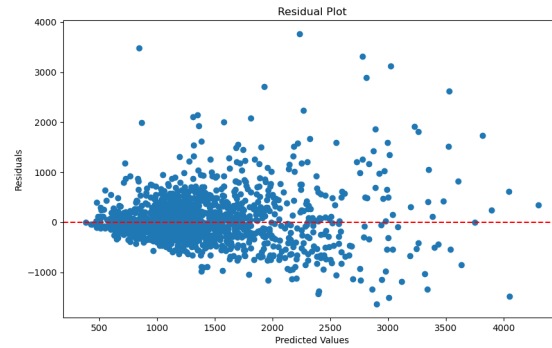


Figure 8: Random Forest residual plot

Multilayer Perceptron Model: After testing six different hidden layer architectures, each with multiple sets of hyperparameters, the model performed best with a shallow network consisting of two hidden layers, with 32 neurons in the first layer and 64 neurons in the second layer. The results were the following: the model achieved a R^2 score of 0.61 on the test and a **RMSE** of 544.5. The residual plots for the model can be seen in Figure 9. These results imply that the MLP model has not captured the nature of the data distribution. The single data point, on the residual plot, where the model overestimates the price by 8000 NOK is especially interesting.

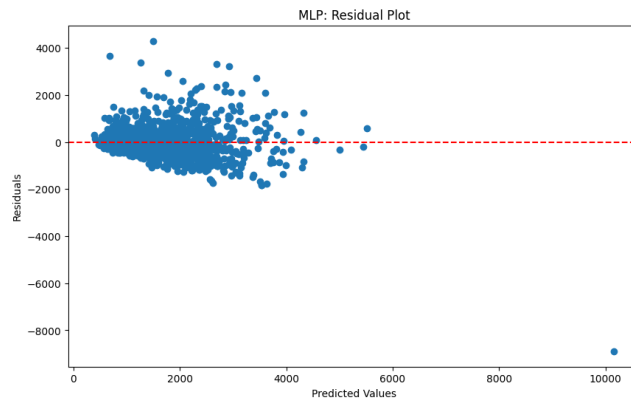


Figure 9: Multilayer Perceptron Residual Plot

Best Model: From these results can be concluded that the Random Forest regressor found the best fitting model. Its performance on the test set is visualised in Figure 10. In this figure it can be observed that for higher prices the predictions are the most inconsistent, while for lower prices the model has better predicting power. This is likely due to a lot more listings having low to medium prices. Sparse data on higher prices leads to a worse fit on that part.



Figure 10: Scatter plot of actual price values against the predicted values (RF model)

Example: A listing has the following properties:

Table 4: An example accommodation.

Feature	Description
Accommodates	2
Bathrooms	1
Bedrooms	1
Longitude	10.74
Latitude	59.92
Host listings count	1
Minimum nights	2
Number of reviews	4
Reviews per month	1.07
Host is superhost	0
Host Identity verified	1
Room type Entire home/apt	1
Number of amenities	545
Distance to Oslo central station	1548.24
Price	1207

The suggested price by the RF model is 1310 NOK, which means that our model is off by 103 NOK.

4.1.3 Objective 3: Determine if the proposed solution performs sufficiently in suggesting appropriate listing prices and fulfills Airbnb's business objectives

The current Random Forest model can account for 65% of the variance. This can be described as a moderately strong predicting model. In its current state, it could thus be able to somewhat aid customers. However, the model performance is deemed as too low for shipping the feature in its current state. Improvements in performance would be needed before shipping. Luckily, the developed model is deemed to have room for improvement. For example the listing description, amenities (the actual written words) and written reviews are not considered in the current model. Incorporating these written words would require significant encoding or the use of a large language model to accurately assess their value. There likely is a lot of 'hidden' information in these parts and Airbnb should explore options to look further into this. Besides exploring new variables, Airbnb could also make use of a gradient boosting algorithm. It also relies on building decision trees, which proved to be the best method in this case study, and, as opposed to the Random Forest model, learns from each tree, likely resulting in a better model fit. This in turn gives better predicting power.

The model would need better performance before shipping because one can easily think of the harmful effects of an under performing model on hosts satisfaction. For example, if the proposed model were to suggest that pricing for a listing was too high it would likely lower booking rate for that listing which could feel unjust by the host. This

could probably be the case with a better performing model as well but would be more justifiable from Airbnb's viewpoint. Therefore, the model would need to perform well enough that it would only suggest that a listing is over or under priced if it was clearly the case. One could also imagine that hosts could try to find ways to exploit the feature by, for example, adding misleading information to their listing in order to make the model suggest that the listing price is lower than it should be. Thereby, making the listing seem more attractive to guests than it actually is.

In conclusion, the proposed solution does not perform sufficiently in its current state and the effects of the low performance could be severe. However, we believe that the performance, illustrated in the results from Section 4.1.2, and the room for improvement, discussed above, indicate that the developed approach could be used to sufficiently solve the problem. Thereby, providing business value for Airbnb.

4.2 Reliability of pipeline

We did automated data quality checks and data pre-processing as described in 3.2. For the feature extraction, we used a correlation matrix to avoid multicollinearity in our features. We made use of k-fold cross-validation to ensure that the model generalizes well and isn't simply overfitting to a particular train/test split. Specific unit tests for individual code components were considered but were deemed as out of scope for our project. Plus, as the functions and methods we apply are very commonly used in the data science community, the chances of unexpected behaviour are reasonably low. Logging of intermediate outputs along the pipeline helps to easily trace back potential errors or irregularities, along with alerts that inform team members in real-time if e.g. data quality issues arise.

4.3 Shortcomings of used method

As mentioned in Section 3.2.2, we decided which features to keep based upon a p -value test in a OLS regression, and a permutation importance test for the Random Forest regressor. Afterwards we used our domain knowledge to determine the final features to keep. We felt that the 76 original features was too many to make a detailed analysis of, within the time span of the project. Therefore, we needed to reduce the amount of features. We needed a clear set of features we could use among all three models, to better compare their results. A major drawback of how we choose to eliminate these features is apparent. Relying on feature selection methods like the p -value test and permutation importance inherently biases our analysis towards features that perform well in linear and tree-based models, respectively. As a result, features that could potentially improve the performance of more complex models, like the Multi-Layer Perceptron (MLP), may have been prematurely discarded. This may have limited the ability of these models to fully capture non-linear relationships present in the data. Furthermore, reducing the feature set to allow comparability among models also likely sacrificed model-specific optimization and should have been avoided.

Additionally, our domain knowledge introduces a subjective element into the feature selection process that is a bit of guessing game. The features we chose may not have

represented the most relevant set for optimizing the prediction accuracy across different models, as it was influenced by our understanding and assumptions about the dataset.

Regarding the shortcomings of a Random Forest regression model, a few key points must be mentioned:

- **Complexity and Interpretability:** Random forest models are often complex and involve numerous decision trees, making them challenging to interpret (Louppe, 2014). Unlike simpler models like linear regression, it is difficult to explain why a specific prediction was made, which can be a drawback when transparency is important.
- **Overfitting Risk:** Although Random Forest reduces overfitting compared to a single decision tree, it can still overfit, especially if the number of trees is too large or the depth of each tree is not controlled effectively. This can limit the model's ability to generalize to new data, particularly when dealing with sparse features.
- **Bias in Feature Importance:** All models can give biased importance to features with more distinct values, especially numeric features, potentially skewing the model's interpretation of what is important in predicting the outcome. Therefore, as some of the feature distributions are quite unstable, this would be likely to happen.

Furthermore, one of the shortcomings of our overarching method was that the decision of problem to focus on was taken based on a limited and possibly unrepresentative sample group. The decision was based solely on the project members opinion and experience with using Airbnb. This could effect the results by implying that the solution fails, or is less successful, in satisfying the business objectives of the larger user population. A better approach could have been to conduct user interviews to see if other users also have the problem of determining if listing prices are appropriate or if this even is the real problem that should be solved.

On the same note, design thinking states that one should exhaustively evaluate all solution alternatives before deciding on a specific option. We tried to do this, but retrospectively find that we were biased towards machine learning alternatives. This can be seen in the results for objective 1, as described in Section 4.1.1, where we tended towards a machine learning approach early on. Machine learning could well be the best solution alternative but it is, nevertheless, a shortcoming of the used method that other options were not evaluated to a greater extent. This is a shortcoming because it could imply that we missed out on solution avenues that could potentially be better than the chosen approach. A better method approach would have been to more systematically evaluate the different solution options in, for example, a brain storming session or workshop after the business understanding and data understanding was completed.

5 Deployment and recommendations

5.1 Implementation plan

As per the project domain the suggested information system prioritizes the platform users with a role of guests. Therefore this action plan will include implementations of interface changes to be altered and added to benefit this user group. The interface will for that reason need a new feature implemented to be able to display the fair market price for an accommodation, and location price history. In addition we recommend extending the same feature for accommodation hosts, in order to share guidance regarding a market appropriate value upon building new listings. In order for a smooth implementation this shall be the second phase of this process, as the underlying system will firstly have to be implemented as detailed in the following implementation plan outlined in Figure 11, where the implementation plan is divided into three distinct phases. In parallel with the introduction of this new feature, we recommend Airbnb initiate a marketing campaign to its user base in order to create awareness of the new platform feature, as this can be key for awareness of the fair market price system.

To ensure a seamless deployment we propose pilot-running the new system to ensure a gradual system load and room for issues to be resolved, this will gradually introduce the system to subsets of users (Stair et al., 2021, p.431). Due to the large user base of Airbnb this will allow for early recognition of potentially arising issues during the implementation process, allowing for user acceptance testing and issue resolutions during its initial introduction phases. This allows Airbnb's users to pilot the project as intended by the project description, allowing Airbnb to gauge the value of this information system during the final phase of its implementation. The final phase of the implementation plan also takes into account tweaks to the new system, to solve or change the system based on feedback and observations, as per a successful value to its customers, tweaks could be preferred over discarding the pilot project. Below the planned implementation is detailed, as per pilot running the start-up of the new system, we recommend starting the system before the interface features are complete, this will allow Airbnb to make necessary changes during its implementation.

Activity	3 Months	1 Month	3+ Months
Implementation	App Integration	UI Integration	Further Development
Marketing		Marketing Campaign	
Start-up		Pilot Running	

Figure 11: Implementation plan

5.2 Recommendations

Throughout this report we outline areas where we can contribute to Airbnb with recommendations regarding the implementation of this new system; in addition to considering a marketing campaign to advertise the new feature. Below we have outlined three key recommendations to Airbnb for what they should do next.

5.2.1 Develop the fair market price system to meet requirements

First, as stated in Section 4.1.3 Objective 3, the model produced during this project does not perform well enough to be implemented right now, and would need to be improved before doing so. We recommend further analysis of the model to enhance its functionality and use cases. As detailed below we recommend building upon the current implementation of appropriate market price system to include spatial data for location specific variables. This will further contribute to an accurate system for Airbnb's users, as well as increasing the overall accuracy of the model. Furthermore, with more capable pre-processing methods and hardware, Airbnb should include more attributes in their implemented model, detailed in the following section regarding limitations to this method and dataset.

Furthermore, as outlined in the final phase of deployment, enhancements may be continuously implemented to increase satisfaction with the system. We recommend putting a focus on this as realistic pricing may differ in a production environment in contrast to the set training data. It is for that reason essential to ensure the system remains functional and adaptable in a real-world scenario. Therefore Airbnb will have to continuously enhance the application to best suit their needs and customer satisfaction, with necessary software enhancements.

Concluding, we recommend that Airbnb continue the project to improve the model which, hopefully, will result in a sufficient feature that can be implemented and maintained in their application.

5.2.2 Implement the new fair market price system to the Airbnb application

As outlined in this report, implementing this new appropriate market pricing system is recommended to create an equitable marketplace for Airbnb's users. This will mitigate accommodation listings that are unfairly priced by suggesting an appropriate price point for both its hosts and guests. Due to this, we recommend implementing this feature, after being improved and validated by Airbnb, to offer its guests an insight into fair and appropriate market pricing.

5.2.3 Marketing campaign

To ensure knowledge of the new platform features, we recommend Airbnb develop a robust marketing strategy to promote them. By increasing its existing users' awareness of the new features this can entice them to return to the platform for their accommodation needs. Such a campaign will contribute to a balanced and competitive marketplace

and increase user understanding of Airbnb’s platform regarding accommodation value, in addition to benefiting other stakeholders.

5.2.4 Stakeholders

With these recommendations Airbnb will see benefit among all its stakeholders. Hosts may see an increase in bookings due to a fair price towards consumers, and consumers will find reasonably priced accommodation suiting their needs. Consequently Airbnb will benefit due to increased traffic to their platform, seeing increased profits to their stakeholders. With continuous development as detailed in the aforementioned section, this system will greatly improve over time as well as increasing customer satisfaction with the platform.

5.3 Limitations of method and dataset

While working on this project, several limitations emerged that related to both the dataset used as well as the methods we employed. These constraints impact the accuracy, but also the generalization of our model. The following outlines the main limitations.

Firstly, our model heavily relies on structured numeric and categorical features from the Inside Airbnb dataset, which lacks potentially valuable, unstructured data. Attributes such as host descriptions, property reviews, and neighborhood summaries, available as text fields, were excluded in this version due to pre-processing limitations and the need for advanced text analysis. However, these data points could provide valuable information regarding perceived property quality, which guests might consider when evaluating listings. Without these variables, the model may overlook other factors that impact price, like property appeal, host responsiveness, and guest feedback.

Secondly, the lack of real-time data constitutes another limitation. The dataset used in this project is only a static snapshot of listings in Oslo from June 29, 2024. But in reality, market conditions and accommodation demand change depending on the season or short-term factors such as events or holidays. If we don’t include time-based data, like occupancy rates, price history, and neighborhood trends, the model’s accuracy might decrease over time. Therefore, implementing dynamic data that pulls real-time listings and historical price changes would certainly improve the model’s adaptability.

Another crucial limitation is the geographical scope of the dataset. While focusing on Oslo allowed us a localized analysis, it also introduces possible biases in feature importance specific to this region. This complicates the possibility to generalize to other cities which would have different accommodation dynamics. While our model takes into account the number of amenities, it does not consider the proximity to specific amenities like grocery stores or tourist attractions. Hence, it may overlook critical price determinants that guests value in certain districts. Integrating geographical features from sources such as OpenStreetMap could help in understanding the impact of local amenities on pricing.

Lastly, while the CRISP-DM framework allowed for a structured approach, the models

that we chose (linear regression, Random Forest, and a multi-layer perceptron) each have their own performance limitations. Linear regression may oversimplify complex interactions between features, while the Random Forest approach, though effective, may not capture complex non-linear relationships. The MLP model, despite offering greater flexibility, struggles with interpretability, which could limit its transparency for users (Zhou et al., 2023). Future iterations might benefit from ensemble approaches or advanced neural architectures to better balance prediction accuracy and interpretability.

5.4 Future analysis

To build on our current results, we suggest several approaches for future analysis that could further extend the model’s capability and accuracy.

One suggestion involves integrating external geographical data through OpenStreetMap (OSM). OSM data can provide detailed information on nearby amenities, distances to city landmarks, public transit hubs, and other points of interest, which guests may value highly when booking their accommodations. By incorporating these spatial features, we can improve the model’s awareness of the respective location and potentially reveal stronger predictors of price variation at a neighborhood level.

Moreover, a temporal analysis integrating seasonality could improve the model’s predictions by considering time-sensitive fluctuations in price. For instance, by capturing historical data patterns and employing time series analysis, the model could predict market changes in advance. This could provide users with context on when prices may rise or fall. This feature would not only add practical value for guests and hosts, but also align more closely with Airbnb’s goals for dynamic pricing.

Another idea is to expand the model to accommodate pricing guidance for hosts. Instead of purely predicting prices, another model which is tailored for hosts can suggest optimal listing prices based on comparable properties, occupancy rates, and dynamic factors such as local event calendars. This would help hosts price their listings competitively, improve customer satisfaction, and keep a good balance between supply and demand in the marketplace.

6 Monitoring and maintenance

In this chapter, we describe our monitoring and maintenance plans. For this, model performance and business KPIs are discussed. Automated alarm triggers for the ‘human in the loop’ are examined and finally, we share our lessons learned.

6.1 Risk analysis

An important aspect that needs continuous attention is data quality. To make sure data quality is reliably high, we implement data quality checks and trigger warnings to the team members in case of increased quality issues. Another risk that we have to consider in the housing-related market is the risk of unintentional biases and reinforcement of existing discrimination against neighborhoods with certain characteristics. We have

to be aware of our potential impact on pricing and the ethical consequences of this. Data privacy is highly valued in Europe. We have to be vigilant of new regulations and always keep the interests of the individual in mind, e.g. when using the listings' location or the text of reviews. These risks can be mitigated by continually engaging with the model and conducting regular sanity checks of the predicted prices.

6.2 Key performance indicators

There are several key performance indicators (KPIs) that are important to keep track of after the product has been implemented. A few are listed down below, but it is likely that unforeseen developments means this will not sufficiently cover all aspects of the product performance. Hence, it is recommended that general reviews of the overall state of the product are carried out periodically.

6.2.1 Model performance

One KPI is the accuracy of price prediction. Since the model is only based on data from a limited time period, it is possible that changing patterns lead to worse performance. To quantify this, the model should be continuously tested on similar datasets as the one used in this project, but for later time periods. The R^2 , **RMSD** and residual plot should be examined, to ensure none of the accuracy measurements trend in the wrong direction.

Other important KPIs related to model performance are crash rate, loading time and availability. If the crash rate is too high or the prediction takes too much time to load, user satisfaction is likely to drop. The percentage of time the price suggestion feature is available is also an important metric to keep track of, and if this is too low, it is both a symptom of frequent errors and a reason for lower usage.

6.2.2 Business KPIs

Many business KPIs to keep track of are related to user behavior. Especially important is quantifying the usage, for example in form of daily active users. The former is normally easily measured as the number of unique users using the product daily. In our case, defining what constitutes a use of the price suggestion feature is not straightforward. If the suggestion is presented together with the price, without requiring any active user interaction, all instances of a user looking at a listing could be considered a use. A more usable definition would be the amount of users that actively choose to, or choose not to book a listing because of the price suggestion. This could for example be quantified by a survey question after a booking has been made, with a question about if the listing was booked and/or other listings disregarded because of the recommendation. Although this will not capture the entire usage, it could be a valuable metric.

Customer satisfaction is also an important KPI. This could be measured by more general survey questions sent out to Airbnb guests, preferably with a different number of bookings.

Financial KPIs relate to the cost of the product, both implementation and maintenance. For this product, these should be related to the set budget decided by Airbnb, and if any steps have a distinctly too high cost, actions should be taken to lower them. These costs should also be related to the overall revenue change since implementing the product.

6.3 Alarms and intervention

In order to guarantee the reliability of the model, a system of automated alarms and intervention protocols is essential. Specific alarms for events will be triggered, for instance when there are sudden, unexplained shifts in pricing recommendations, consistent errors in demand forecasting, or unusual booking behaviours like abrupt declines in bookings for certain listings. When these triggers are activated, the system should send an alarm so that a human can assess the flagged instances and make the necessary corrections.

This approach follows the “human in the loop“-model, where the responsible team members, such as operations staff or data scientists, are alerted to oversee the system rather than intervene directly with every flagged instance (Fischer et al., 2017). For example, if a sudden jump in demand predictions causes notable price increases that don’t align with the actual trends, an alarm is sent to the respective responsible. That human can subsequently investigate potential issues like data quality concerns, model drift, or temporary market anomalies (such as local events). Similarly, if pricing recommendations start to differ significantly from competitor rates or similar local listings, an alert could signal a review to check for possible biases or errors.

6.4 Lessons learned

Booking meeting room

We learned that booking a meeting room several days (if not weeks) prior to a group meeting is key for a successful project.

The relevance of CRISP-DM

Through working with the project we got hands on experience conducting a (almost) full fledged data science project. In doing so, the relevance of the CRISP-DM framework became apparent. The framework provided a clear framework to follow and also highlights the importance of the project serving a business purpose.

Importance of design thinking

Similar to the previous lesson learned, by working through the life cycle of a data science project, the importance of design thinking and solving the correct problem became clear. We followed the CRISP-DM framework and tried to internalize the principles of design thinking, but we realized that, even then, it is easy to get influenced by bias and jump right to solving a problem. Using more time and more systematically examining the problem during the business understanding to find a real and wide spread problem would probably have been valuable.

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